PHYSICS-INFORMED NEURAL NETWORKS (PINN) TO SOLVE KINEMATIC WAVE EQUATION (KWE) FOR FLOOD ROUTING

ABSTRACT

Floods, among the most destructive natural disasters, demand accurate modelling of flood wave propagation to enable effective management, prediction, and mitigation. The Kinematic Wave Equation (KWE) is a critical hydrological model for simulating flood routing, yet traditional numerical methods like the Finite Difference Method are computationally intensive and require detailed data, often scarce in real-world scenarios. This study proposes a Physics-Informed Neural Network (PINN) approach to solve the KWE, embedding physical laws into the neural network's loss function to deliver physically consistent solutions even with limited data. The objectives are to develop a PINN model for both linear and non-linear KWE forms, validate its accuracy against known numerical solutions, and optimize hyperparameters, such as network depth, neurons, learning rate, and activation functions, to enhance convergence and performance. The study focuses on a one-dimensional flood routing problem with a sinusoidal inflow pattern, implemented using PyTorch's automatic differentiation for computational efficiency. The project's feasibility is supported by modern computational resources, established neural network theories for solving differential equations, and Python's robust machine learning ecosystem. By integrating data-driven and physics-based approaches, this research aims to provide a scalable, accurate alternative to traditional methods for flood routing predictions in normalized river systems.

INTRODUCTION

Floods cause significant economic, environmental, and human losses, necessitating accurate flood wave propagation models for effective management. The Kinematic Wave Equation (KWE) simulates flood routing, but traditional methods like FDM are computationally intensive and data-dependent. Physics-Informed Neural Networks (PINNs) embed KWE's physical laws into neural networks, offering a mesh-free, data-efficient alternative for linear and non-linear flood dynamics (Raissi et al., 2019)

PROBLEM OF STATEMENT

- Existing flood modelling methods (e.g., Kinematic Wave Equation with numerical approaches) are computationally intensive and time-consuming.
- Traditional models require extensive data and struggle to accurately represent both linear and non-linear flood behaviour.
- Current approaches face challenges in maintaining accuracy, physical consistency, and scalability for real-world applications.

OBJECTIVES

- Develop a PINN model to solve linear and non-linear KWE for flood routing.
- Validate PINN accuracy against FDM solutions.
- Optimize hyperparameters (e.g., layers, neurons, learning rate) for convergence.

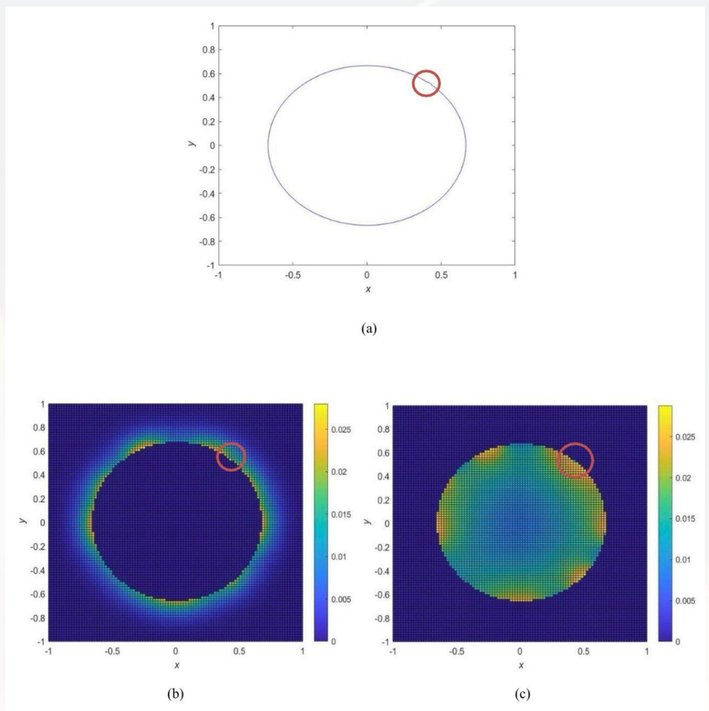
SCOPE OF STUDY

- Developing and apply an alternative modelling approach for flood wave propagation based on the Kinematic Wave Equation framework.
- Analysis of both linear and non-linear flood routing conditions using selected input parameters and simplified datasets.
- Evaluated in terms of computational efficiency, accuracy, and physical consistency through comparison with conventional numerical methods.

METHODOLOGY

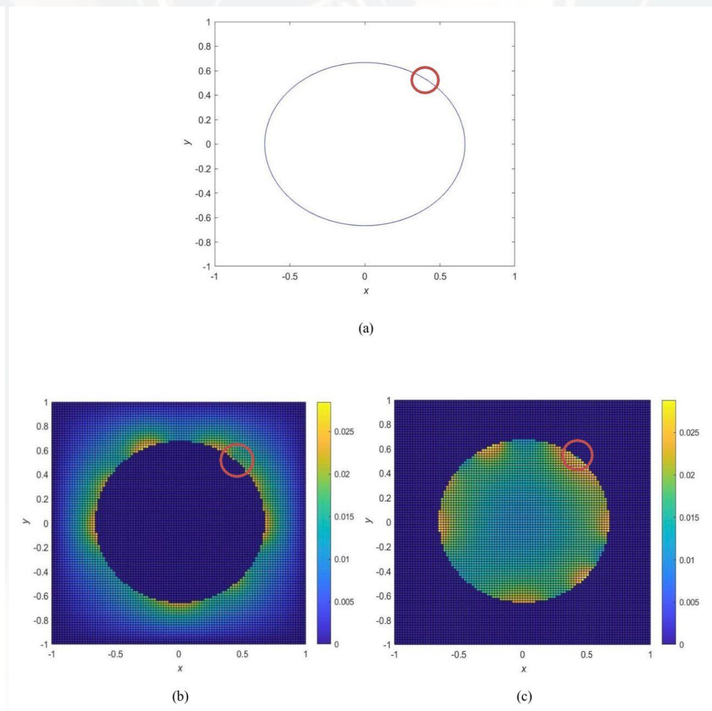
This study develops a PINN model to solve the KWE for flood routing using PyTorch's automatic differentiation. The model addresses a one-dimensional problem with sinusoidal inflow, embedding KWE's PDEs, initial conditions (e.g., , and boundary conditions into the loss function. Hyperparameters (2–4 layers, 20–50 neurons, learning rates 10^{-2} to 10^{-4}, tanh/Sigmoid/ReLU activations) are tuned to optimize accuracy and convergence.

RESULTS & DISCUSSION



(a) (b) (c)

The results for (a) interface position (psi), (b) ligand density, and (c) signal density for $d_{c^*} = 0.1$



(a) (b) (c)

The results for (a) interface position (psi), (b) ligand density, and (c) signal density for $d_{c^*} = 100$

Comparison of Results

Parameter	Low Diffusion (0.1)	High Diffusion (100)
Gradient	Sharp	Smooth
Protrusion	Early formation	No formation
Stability	Good	Poor

CONCLUSION

1. Mathematical model successfully captures ligand diffusion, signal transduction and membrane deformation
2. Simulation results show ligand diffusion rate significantly affects invadopodia formation and sharp gradients promote protrusion
3. Contribution: Enhances understanding of cancer invasion mechanisms and offers potential for future drug development targeting ligand dynamics

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Rabiatul Adawiyah Supa'at
Universiti Teknologi Malaysia
rabiatul02@graduate.utm.my



Dr. Noorehan Yaacob
Universiti Teknologi Malaysia
noorehan@utm.my